

Student Placement Project

Problem definition:

The problem addressed is predicting whether a student will be placed or not based on academic and behavioral factors such as study hours, attendance, sleep hours, internet usage, assignments completed, previous scores, and exam scores.

Objective of Project :

The project is to develop a ML model that can accurately predict a student's placement status using historical student data. This helps in identifying student's placement chances and understanding the factors that influence placement outcomes.

Dataset Description :

The dataset contains information about students, including academic performance and study-related habits, which are used to predict the placement status.

Features (Input Variables):

- Study hours
- Attendance
- Sleep hours
- Internet usage
- Assignments completed
- Previous score
- Exam score

Target Variable:

- Placement Status: Indicates whether a student is placed or not.

Model Justification:

Linear Regression was selected because it can learn the relationship b/w student performance factors and placement status. All the data is in numerical form also. It is simple and suitable for understanding how input features influence prediction.

Conclusion:

The project successfully developed a ML model to predict student placement status based on academic and behavioral factors. The model was able to make predictions using the provided student data.

Limitations:

The model sometimes produces slight prediction errors, requiring rounding to the output to obtain valid placement status value.

Future improvements:

The model's accuracy can be improved by using a larger dataset and testing more advanced classification algorithms such as logistic regression, Decision trees or Random Forests.

Project Reflection

Mini Project - Student Placement

Through this project, I understood how ML can be used to predict a student's placement status based on factors such as study hours, attendance, sleep hours, internet usage, assignments completed, previous score and exam score. I learned how raw data can be transformed into predictions using a trained model.

One of the main challenges I faced during implementation was handling the slight prediction error produced by the Linear regression model. Since the target variable was encoded as 0 or 1, the model sometimes returned values slightly above or below these numbers, such as 1.0019. To obtain the correct classification result, I had to apply rounding to the prediction output. I also had to ensure that preprocessing steps such as label encoding and scaling were applied correctly.

The key concepts I learned includes data preprocessing, label encoding, Feature scaling using Standard Scaler, train-test splitting, model training with Linear Regression, model evaluation using Mean Squared error and RMSE and model persistence using Joblib. I also learned how to load a saved model and use it to make predictions on new data.

The project helped me to understand the complete ML workflow, starting from data collection from 'Kaggle' and cleaning, followed by preprocessing, model training, evaluation, saving the model, and finally making predictions on unseen data. It provided practical experience with each stage of the ML pipeline and improved my understanding of how ML models are developed and deployed for real-world applications.